Stormwater Guidelines for Development



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Bega Valley Shire Council acknowledges and pays our respects to the traditional custodians of the lands, waterways and airspace of the shire.

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List of amendments

Revision	Date adopted by Council	Date in effect	Amendments
0 - Initial	15 July 2026	15 September 2026	N/A

Abbreviations and Definitions

0074 Stormwater Drainage (Design)	Council's specification for stormwater drainage design
AEP	Annual Exceedance Probability
AR&R	Australian Rainfall and Runoff: A Guide to Flood Estimation 2019
AS/ NZS 3500.3	Australian Standard for stormwater drainage
Chartered Professional Engineer	An engineer accredited by Engineers Australia who has formally met the Stage 2 Competency Standards. Accreditation is in a particular discipline or disciplines
Council	Bega Valley Shire Council
DA	Development Application
DCP 2013	Bega Valley Development Control Plan 2013
Detailed Design	Detailed stormwater design submitted with either the Subdivision Works Certificate (SWC) or Section 68 Application or Section 138 application
Easement to Drain Water	As defined in the *Conveyancing Act 1919*
Infill subdivision	means subdivision that does not involve the creation or extension of roads
Infill development	means building on, or other development of previously subdivided residential, commercial, industrial or rural residential land, and generally does not involve the creation of a new public road
LEP 2013	*Bega Valley Local Environmental Plan 2013*
Major subdivision	means subdivision of land, whether for the purpose of urban, village, commercial, industrial or rural residential development, that includes the creation of new roads
OLFP	Overland Flow Path
OSD	On-Site Detention
PSD	Permitted Site Discharge
Professional engineer	A person who: - Holds a four-year professional engineering degree or equivalent that is accredited by Engineers Australia, and has appropriate experience and competence in the relevant field, OR - Holds a post-graduate or master's degree that is accredited by Engineers Australia and has appropriate experience and competence in the relevant field.
RC	Reinforced Concrete
RCP	Reinforced Concrete Pipe
Registered Engineer	A person who is:

	- A *professional engineer* who is registered on the National Engineering Register (NER) - A *professional engineer* who is registered to work on Regulated Buildings including ancillary works under the NSW Design and Building Practitioners (DBP) scheme
Regulated building	Class 2, 3 or 9c building or a building that includes a class 2, 3 or 9c part as classified under the National Construction Code (NCC)
Roads Authority	Council, Transport for New South Wales or Crown Lands
Section 138 Application	Application under Section 138 of the *Roads Act 1993* to undertake works in the Road Reserve. Application is made to the relevant Roads Authority
Section 68 Application	Application under Section 68 of the *Local Government Act 1993* to connect to or alter Council's stormwater drainage system
Section 88B instrument	Instrument relating to an easement, profit à prendre, or restriction on use of land under Section 88B of the *Conveyancing Act 1919*
SQID	Stormwater Quality Improvement Device
SWC	Subdivision Works Certificate
TWL	Top water level
Waterway and Watercourse, waterbody, wetland, estuary	As defined in the standard instrument, principal Local Environmental Plan (LEP 2013) See section 2.1

1 Introduction

These guidelines supplement Council's Development Control Plan. They prescribe Council's requirements for the management of stormwater within development sites and discharge of stormwater from development sites. They must be used in conjunction with:

- Council's Development Design and Construction Specifications
- Council's Development Control Plan (DCP) 2013
- AS/NZS 3500.3 - Plumbing and Drainage Part 3: Stormwater drainage

Stormwater objectives for development in the Bega Valley Shire are:

- Pre-development water quality is maintained or enhanced during construction and in post-development runoff.
- To address the impact of decreasingly pervious catchments on downstream property, infrastructure and the environment.
- To incorporate the principles of water sensitive urban design and integrated water cycle management.

Document format

The format of these guidelines is informed and adapted from the Australian Model Code of Residential Development (AMCORD). Each sub-section generally follows the below format:

- ***Application:*** the application details what zones or development types the section applies to.
- ***Objectives:*** the objectives are a statement outlining the desirable planning and development outcome/s. The objectives can be achieved by either applying the identified acceptable solutions, or by preparing a performance-based solution.
- ***Explanation:*** the explanation assists in the interpretation of the objectives, performance criteria and acceptable solutions. It provides background information and explanatory evidence for the performance criteria, together with examples and illustrations to clarify principles and demonstrate 'best practice'. It may provide clarification on the priority of development controls for instances where it is not possible to comply with competing objectives or other more prescriptive development controls.
- ***Performance criteria:*** the performance criteria detail the outcomes required to achieve the objective/s. They are written to provide the maximum amount of flexibility in achieving the objective***.*** Where the designer nominates to address the performance criteria with a performance-based solution, the application will be assessed against the performance criteria and relevant objectives.
- ***Acceptable solutions:*** the acceptable solutions provide a prescriptive solution for addressing the objectives which Council will accept without requiring further justification or explanation.

  Where an Acceptable Solution satisfies all the aspects of a Performance Criterion, it is given the same number. In these cases, designers have the choice of satisfying the Performance Criterion using an alternative solution or conforming to the Acceptable Solution. E.g.,

  P1 _____ A1 _____

  Where more than one Acceptable Solution combine to satisfy a Performance Criterion, the following format is used:

  P1 _____ A1.1 _____

  AND

A1.2 _____

Where there is more than one Acceptable Solution that satisfies a Performance Criterion, the following format is used:

P1 _____ A1.1 _____

OR

A1.2 _____

- ***Resources:*** A list of resources that may assist in meeting the objectives or performance criteria.

The Stormwater Report or Statement of Environmental Effects must include a response to either the Performance criteria or Acceptable solution for each applicable row in the tables, demonstrating how the requirement has been met.

1.1 Lodgement and approvals

Application

Development type	All development
Zones	All zones

1.1.1 Development application stage

Stormwater infrastructure in public land not owned by Council

Where stormwater infrastructure is proposed to traverse or discharge into land owned by an authority other than Council (e.g. Crown, TfNSW), the applicant must obtain landowners consent for lodgement of the DA from that authority.

Stormwater Plan

All Development Applications (DAs) must be accompanied by a stormwater plan, except:

- Where there is no change to roof or hardstand areas (e.g. internal alterations and additions, home businesses without new buildings),
- The development is a low impact rural land use that will generate negligible additional run-off or result in negligible ground disturbance (e.g. bee keeping, roadside stalls),
- The development is a cemetery or home burial, or
- When a **stormwater concept design and report** is required instead (see below).

The stormwater plan must illustrate full drainage paths from roofs and hardstand areas to Council's stormwater system / or a drainage easement.

The stormwater plan can be completed by the builder/designer/architect/drafter.

Stormwater concept design and report

Development Applications (DAs) must be accompanied by a stormwater concept design and report prepared by a *professional engineer* under the following circumstances:

Circumstance	Exceptions
Stormwater constrained sites	Single dwellings with more than 1.5m head between the lowest outlet and kerb/ street outlet

Residential development	Single dwellings, secondary dwellings, residential ancillary development, alterations and additions where the increase in impervious area is ≤70m2 and other urban residential developments containing less than 7 units and where post development impervious fraction is ≤40%
Development located 1km up the catchment, or within 1km of any associated estuary that feeds into an existing oyster aquaculture development or priority oyster aquaculture area (as defined in *State Environmental Planning Policy (Primary Production) 2021*)	
Development located 1km up the catchment of a wetland	
Subdivisions	Boundary adjustments
Where public stormwater works are proposed	Nil
Commercial developments	Where there is no change to roof or hardstand areas
Where proposing or impacting overland flow paths	Nil

Reporting requirements are provided in Section 6.1.

Other development types

Any development types or circumstances that do not fit above will be assessed on a case-by-case basis considering the development scale and downstream stormwater impact.

1.1.2 Construction stage

Construction stage – stormwater connections via Section 68 approval

An approval under Section 68 of the *Local Government Act 1993* must be obtained for construction of stormwater connections to Council's network. For basic connections where no stormwater concept design and report was required at DA stage, the connection details can be provided on the architectural plans in conformity with these guidelines. This approval is typically a combined approval with the development's sewer connection.

Where stormwater concept design and report was required at DA stage, engineering details of the proposed connection must be provided.

For specialised connections a separate Section 68 application accompanied by engineering details of the proposed connection will be required as indicated by the development consent.

Construction stage – Private works

Complex private drainage systems, on-site detention and Stormwater Quality Improvement Devices (SQID)s are required to be designed by a *professional engineer* in accordance with these guidelines and Australian standards prior to the issue of a construction certificate for the development.

Detailed design, construction and certification requirements and timing are stipulated through conditions of development consent and/or as conditions of a Section 68 or Section 138 approval. Evidence of compliance with these requirements is typically required to be provided to the principal certifier.

Construction stage – Public Works

A Section 68 approval must be obtained for construction or upgrade of Council's network where it is within private property.

An approval under Section 138 of the *Roads Act 1993* is required for all public stormwater works within public roads.

A Subdivision Works Certificate (SWC) is required for new stormwater infrastructure to be constructed as part of subdivision works.

A design report must accompany all submissions.

Timing as indicated by the development consent. See section 6.2 for design submission requirements. Design to be in accordance with these guidelines and Council Specification 0074 - *Stormwater drainage (Design).*

1.1.3 Complying Development Certificates

See DCP 2013 Section 5.14 for complying development types that require Section 68 approval prior to the issue of a Complying Development Certificate (CDC). For these development types, submission requirements include:

- Architectural site plans showing internal stormwater systems.
- Stormwater design documentation in accordance with Section 6.1.

For other complying development, the Section 68 submission requirements are as per *Construction stage – stormwater connections via Section 68 approval* in Section 1.1.2 of this document.

1.1.4 Prior to occupation or subdivision certificate

The following is a summary of the stormwater documentation required to be submitted prior to occupation for building works or subdivision certificate for subdivision works. Additional documentation may be required, or timing varied on a case-by-case basis as indicated by the development consent.

Private works

Prior to the issue of an occupation certificate for building works the Principal Certifier must be provided with:

- Certification of private OSD and/or SQID systems as per section 6.5.
- Evidence that the OSD system has been registered on the property title in accordance with section 6.6.
- Any other certification of private engineering works associated with stormwater such as retaining walls, culverts etc, typically as required by the development consent.

Public works

Prior to the issue of an occupation certificate or issue of a subdivision certificate, Council must be provided with:

- Works as executed plans for public or inter-allotment drainage works in accordance with section 6.3.
- CCTV report of any new Council stormwater and evidence of completion of any remediation work as per section 6.4.
- Certification of public OSD and/or SQID systems as per section 6.5.
- Any other certification of approved public engineering works associated with stormwater such as retaining walls, culverts etc, as required by the development consent.

2 Stormwater Disposal

2.1 Legal point of discharge

Application

Development type	All development
Zones	All zones

Objective

- Stormwater is discharged from development sites via a legal point of discharge.

Performance Criteria	Acceptable Solutions
The objectives may be achieved where:	
P1 *Acceptable solutions A1.1 - 1.5 are the only options acceptable to Council.* **Notes:** • *Waterway* is a defined term in the Standard Instrument – Principal Local Environmental Plan, and means the whole or any part of a watercourse, wetland, waterbody (artificial) or waterbody (natural). For the purpose of this guideline includes a minor stream and higher order streams as defined in the *Water Management (General) Regulation 2025* if shown on LEP 2013 mapping as a 'blue line' watercourse. • Section 3 contains provisions for on-site disposal of stormwater for certain low-density developments on suitable land.	A1.1 Stormwater is discharged through conveyance infrastructure under the lawful control of Council that shares a common boundary with the subject lot or extends into the subject lot. OR A1.2 Stormwater is discharged through conveyance infrastructure within private land, with an appropriate stormwater easement favouring the subject lot, that connects to a Council system or *waterway*. OR A1.3 Stormwater is discharged into a *waterway* to which the development site naturally drains. OR A1.4 Stormwater is discharged over rural land in accordance with the additional provisions in section 2.3.2. OR A1.5 Stormwater is discharged into bushland but only where the provisions in section 2.2.5 are satisfied.

Resources

Standard Instrument – Principle Local Environmental Plan

2.2 Urban, village, commercial and industrial settings

Application

Development type	All development
Zones	R2, R3, RU5, E1-E5, MU1, SP1-SP3, RE1, RE2

Objectives

- Stormwater is discharged into appropriate receiving infrastructure.
- Stormwater discharge into public land is controlled and permissible.

2.2.1 Disposal hierarchy

Explanation

This clause sets out the order of preference for stormwater disposal method. Further details are provided in sub sections 2.2.2-2.2.7.

Note: for Stormwater constrained sites, where properties fall naturally away from the street and there is no Council or inter-allotment drainage system available, see Section 3.

Performance Criteria	Acceptable Solutions
The objectives may be achieved where:	
P1 *Acceptable solution A1 is the only option acceptable to Council.*	A1 The method of stormwater disposal is selected based on the following order of preference: 1. Public Roads, OR Conveyance systems on private property covered by an appropriate easement (Council or inter-allotment), OR Conveyance systems within a drainage reserve, OR A *waterway*. 2. Stormwater discharge to bushland (where permitted), 3. Piping stormwater through Council owned and managed land (where permitted).

2.2.2 Stormwater connection to public roads

Performance Criteria	Acceptable Solutions
The objectives may be achieved where:	
P1 *Acceptable solution A1 is the only option acceptable to Council.*	A1 Where stormwater is proposed to be disposed into a public road, the method of stormwater disposal is selected based on the following order of preference: 1. Discharge directly into an existing Council pit where available within 10m of the lot frontage and falls permit, 2. Discharge to kerb where permitted in accordance with 2.2.2.1, 3. Connect to the obvert of a Council stormwater pipe, 4. Discharge directly into the table drain where no kerb and gutter or Council stormwater pipe exists. a) R3 and MU1 zones - only permitted for single dwellings and ancillary development b) R2 zones – only permitted for up to 3 dwellings on a lot or 1 into 2 lot subdivisions.

2.2.2.1 Kerb connections

Performance Criteria	Acceptable Solutions
The objectives may be achieved where:	
P1 Kerb connections do not result in degradation of the kerb.	A1 Connection to the kerb is via a suitably manufactured galvanised steel adaptor that matches the kerb profile.
P2 Discharge does not exceed 30 l/s per 15m of kerb for a single pipe, with a total maximum discharge of 60 l/s.	A2 For single residential development, alterations and additions, and dual occupancy kerb outlets are provided at a rate of 1 x 90mm kerb outlet per 250m2 of impervious area. A maximum of two outlets is permitted.

2.2.3 Inter-allotment drainage and public drainage systems on private property

Explanation

Stormwater discharged through private property via private systems must have easements in place. See Section 2.4 for detail on the timing and particulars of easement creation.

In subdivisions, a balance needs to be struck between having an excessive number of small inter-allotment drainage systems and simplifying maintenance and for the users.

New back of lot Council drainage systems are not accepted owing to the difficulties in gaining access to and maintaining these systems.

Performance Criteria	Acceptable Solutions
The objectives may be achieved where:	
P1 Stormwater is conveyed across private land via Inter-allotment drainage systems.	A1.1 Inter-allotment drainage systems are designed in accordance with Councils Specification 0074 – *Stormwater drainage (Design)*. AND A1.2 An easement to drain water is created in favour of the lot(s) discharging stormwater over all burdened lot(s). **Note:** See Section 2.4 for particulars.
P2 New inter-allotment drainage systems minimise the potential for maintenance conflicts between landowners.	A2 Any new inter-allotment drainage system serves no more than four lots.
P3 Council does not have liability for maintaining new stormwater assets on private land.	A3 New stormwater drainage systems within private land are private inter-allotment drainage systems, **Note:** Council will not accept new back of lot public stormwater systems.
P4 Stormwater discharge into existing inter-allotment drainage systems is controlled.	A4.1 Stormwater discharge from the development is directed into an existing stormwater pit within the inter-allotment drainage system. OR A4.2 A new pit is installed within the inter-allotment drainage system, and stormwater is directed into that pit.
P5 Stormwater discharge into existing back of lot Council systems is controlled.	A5 Stormwater discharge from the development is directed into a boundary pit which is connected into a pit within the Council drainage system.

2.2.4 Connection to stormwater channel or waterway

Explanation

When connecting into a stormwater channel or *waterway* the developer must ensure that the impacts to the environment, amenity and maintenance of public spaces is minimised.

Waterway is a defined term given in Section 2.1.

Performance Criteria	Acceptable Solutions
The objectives may be achieved where:	
P1 Connections into a stormwater channel or *waterway* protect against erosion and the number of connections is minimised to promote amenity and minimise maintenance issues for the land manager.	A1.1 Connections are made into an existing piped connection, if one is available near the development site. If one is not available, then the connection is made via a single connection point to the channel or *waterway* adjacent to the site. AND A1.2 a) If the channel is lined, the developer reinstates the lining of the channel at the connection point, to Council's satisfaction. b) If the channel or *waterway* is unlined, the channel/ *waterway* is protected against erosion at the point of discharge. The protection must be in the form of a durable rock apron, to Council's satisfaction. **Note:** For rural situations, a standard design is provided in Appendix F.

2.2.5 Stormwater discharge to bushland

Explanation

Proposals to discharge stormwater into Council owned bushland will be assessed on a case-by-case basis.

The developer is encouraged to consult with Council's property services team prior to submitting a DA that relies on this disposal type.

Where bushland is not owned by Council the applicant must apply to and obtain written approval from the relevant authority that controls the bushland prior to lodging any DA for development requiring stormwater drainage works.

Performance Criteria	Acceptable Solutions
The objectives may be achieved where:	
P1 Council is satisfied that the proposal will not contravene the land classification and will not adversely impact the maintenance, amenity, recreational and environmental values of the land.	A1.1 Stormwater is discharged into Council owned bushland only where: The classification of the land under Section 26 of the *Local Government Act 1993* is 'operational'. AND A1.2 It is demonstrated and agreed to by Council that the proposal can be accommodated within any plan of management, maintenance regime or covenant that applies to the land. AND A1.3 The proposal demonstrates it can avoid sterilising any future use of the land, based on the current land zoning. AND A1.4 The design incorporates pipe outlet energy dissipaters to reduce energy and to spread or direct the flow in a uniform manner, as the location requires.
P2 *Acceptable solutions A2.1 - A2.2 are the only options acceptable to Council.*	A2.1 Stormwater is discharged into non-Council owned bushland only where the developer has obtained written approval from the relevant authority that controls the bushland. AND A2.2 The design incorporates pipe outlet energy dissipaters to reduce energy and to spread or direct the flow in a uniform manner, as the location requires.

2.2.6 Piping stormwater drainage through Council owned and managed land

Explanation

Proposals to construct stormwater drainage and discharge infrastructure through Council owned and managed land will be assessed on a case-by-case basis.

The developer is encouraged to consult with Council's property services team prior to submitting a DA that relies on this disposal type.

Performance Criteria	Acceptable Solutions
The objectives may be achieved where:	
P1 Council is satisfied that the proposal will not contravene the land classification and will not adversely impact the maintenance, amenity, recreational and environmental values of the land.	A1.1 Stormwater is discharged into Council owned and managed land only where: The classification of the land under Section 26 of the *Local Government Act 1993* is 'operational'. AND A1.2 It is demonstrated and agreed to by Council that the proposal can be accommodated within any plan of management, maintenance regime or covenant that applies to the land. AND A1.3 The proposal demonstrates it can avoid sterilising any future use of the land, based on the current land zoning. AND A1.4 The minimum drainage structure crossing public land is 375 mm RCP. Erosion protection and/or energy dissipators are provided at the point of discharge. **Notes:** Council may require the drainage structure to be sized to allow for future connections from adjoining properties. Council may seek compensation for granting easements and construction of stormwater drainage infrastructure through publicly owned land. The developer must bear all costs.

2.2.7 Extending Council's stormwater system

Explanation

Where extensions to Council's stormwater system are approved to facilitate a development, it is the developer's responsibility to undertake all acquisitions, obtain approvals from Council and complete all construction works required to extend the Council system.

Council stormwater systems must be designed in accordance with Council Specification 0074 - *Stormwater drainage (Design).*

Performance Criteria	Acceptable Solutions
The objectives may be achieved where:	
P1 Council stormwater systems are extended to facilitate development in the following circumstances: a) Where the minor and/ or major system flows from the development cannot be accommodated by the existing point(s) of discharge and their downstream systems, or b) As part of subdivisions that are creating public assets (except new back of lot public stormwater), or c) In conjunction with road upgrades, or d) For infill development, where no point of discharge that complies with the requirements of section 2.2 or section 3 is available and extending Council's existing system is the only feasible way for the development to progress.	A1 *There are no recommended Acceptable Solutions for this Element, as each situation requires an individual approach.*
P2 *Acceptable solution A2 is the only option acceptable to Council.*	A2 All costs associated with extending Councils stormwater system are borne by the developer except where a VPA is entered into with Council.

2.3 Large lot and rural stormwater disposal

Application

Development type	All development
Zones	RU1, RU2, RU4, C3, C4, R5

Objectives

- Stormwater discharge does not cause nuisance or damage to neighbouring land.
- Stormwater discharge does not adversely impact the environment.

Explanation

Where development is a long way from downstream boundaries, localised erosion control along with sensible design will generally provide adequate protection against environmental damage and property damage.

Were development is close to downstream boundaries. Council must be satisfied that a legal point of discharge is available and an acceptable mechanism for discharge is provided.

2.3.1 Development greater than 50m from a downstream boundary except subdivision

Performance Criteria	Acceptable Solutions
The objectives may be achieved where:	
P1 Stormwater discharge is controlled so as not to: • cause nuisance to neighbours, or • flow through wastewater systems, or • cause erosion.	A1 The development plan(s) demonstrate that stormwater discharge is not directed towards on-site wastewater systems and that erosion control is provided. **Note:** A standard stormwater energy dissipater layout for a single dwelling is provided in Appendix F.

2.3.2 Development within 50m of a downstream boundary and all subdivisions

Performance Criteria	Acceptable Solutions
The objectives may be achieved where:	
P1 *Acceptable solutions A1.1 - 1.6 are the only options acceptable to Council.*	A1.1 Stormwater is discharged via a legal point of discharge as defined in section 2.1. OR A1.2 Where the lot size(s) ≥ 1 Ha, stormwater is discharged into a *waterway* directly adjacent to the subject lot if the top of the nearest bank (or centreline where there are no banks) is within 20m of the property boundary and there is no existing development between the boundary and the *waterway*. OR A1.3 Where the lot size(s) ≥ 1 Ha, stormwater is discharged into a gully (defined by survey or topographic mapping) within the subject lot that is not mapped as a *waterway* but directly intersects a *waterway*. OR A1.4 Where lot sizes < 1 Ha, Stormwater is discharged into a *waterway* directly adjacent to the lot if the top of the nearest bank (or centreline where there are no banks) is within 10m of the property boundary and there is no existing

	development between the boundary and the *waterway*. OR A1.5 Stormwater is discharged via a downstream easement. OR A1.6 Stormwater is discharged via a system that limits 1% AEP flows to pre-development levels and spreads flows. **Note:** The system must be designed by a professional engineer, except in the case of a single dwelling where the standard stormwater energy dissipater layout provided in Appendix F will be accepted.
P2 Stormwater discharge is controlled so as not to: • cause nuisance to neighbours, or • flow through wastewater systems, or • cause erosion.	A2 *There are no recommended Acceptable Solutions for this Element, as each situation requires an individual approach.*

2.4 Drainage Easements

Application

Development type	All development
Zones	All zones

Objectives

- All landowners and stakeholders understand their rights and responsibilities in relation to the creation and operation of easements to drain water.
- Easements are fit for purpose.

Explanation

This section covers the mechanisms for creation, timing and width of easements. For details on discharge of stormwater through private land see Section 2.2.3.

Performance Criteria	Acceptable Solutions
The objectives may be achieved where:	
P1 *Acceptable solution A1 is the only option acceptable to Council.*	A1.1 For development other than subdivisions, the landowners' written agreement for the creation of any easement to drain water

			benefitting the development is provided as part of the development application. AND A1.2 Documents creating the easement to drain water are lodged and registered with Land Registry Services (LRS) prior to the development consent becoming operational. I.e., the development consent may be issued as a Deferred Commencement Consent, such that the registering of the easement is required for the consent to operate.
P2	*Acceptable solution A2 is the only option acceptable to Council.*	A2	For subdivisions, any downstream lot to be burdened by an easement to drain water is party to the development application and the appropriate landowners consent provided as part of DA lodgement. **Note:** Council will require the easement be detailed on the plan of subdivision and associated Section 88B instruments lodged with Council at subdivision certificate stage, and subsequently with NSW Land Registry Services.
P3	*Acceptable solution A3 is the only option acceptable to Council.*	A3	Where an existing Council stormwater system not covered by an easement favouring Council is either; • Traversing the site, or • Proposed to be connected to by the development An easement favouring Council must be created by the applicant, in accordance with the *Conveyancing Act 1919,* prior to the issue of an occupation certificate or subdivision certificate. All costs for the creation of easements must be borne by the developer.
P4	The width of easement(s) to drain water allow for construction, operation and maintenance of the stormwater infrastructure.	A4	Easement widths are as per Table 1.
P5	*Acceptable solution A5 is the only option acceptable to Council.*	A5	All easements must be accompanied by a suitable instrument (typically 88B) setting out any terms and maintenance rights and responsibilities that are not covered in, or that vary from, the standard instrument.

Table 1 - Minimum Easement Widths

Pipe Diameter (mm)	Minimum Width of Easement to Drain Water (m) [1]
100, 150, 225 & 300	1.0[2]
375 - 900	3.0
1050	3.5
1200, 1350	4.0
1500, 1650, 1800	5.0
Multiples	External width of the laid pipes plus 2 times diameter. Min 3m
An easement benefitting Council	Minimum 3m wide, otherwise as above
Overland flow path	The width defined by the major 1% AEP event including climate change factors where appropriate.

1. Where the depth of pipes necessitates benching to install/ maintain, wider easements to suit may be necessary. This is to be determined by the designer (professional engineer).
2. The easement width may be reduced to 0.9m between existing dwellings and boundary.

2.5 Overland flow

Application

Development type	All development
Zones	All zones

Objectives

- Development impacting, re-directing or creating overland flow does not result in adverse impacts to surrounding properties, safety outcomes or the environment.
- As far as practical, ensure new development does not increase risk to life.

Explanation

Overland flow paths collect local run-off and drainage system surcharge flows and convey them to *waterways.* Overland flow paths may be created as part of a development or impacted by a development. Development Application considerations for overland flow paths are covered in more detail in Section 5.8 – *Planning for Hazards* of Councils DCP.

Sub-catchment redirection of stormwater may increase flows beyond what the receiving system was designed for and increase chances of flooding of an Overland Flow Path. Wherever sub-catchment re-direction is proposed it must be supported by detailed calculations and analysis at DA stage.

Developments must make provision for existing upstream overland flows through the site. Care must be exercised to ensure that the provision of such diversions within the site does not result in the concentration of stormwater onto adjoining properties. If adjacent properties are

impacted, then a suitable design that may include upsizing the OLFP, detention and/ or dissipation of stormwater inflows is required.

Performance Criteria	Acceptable Solutions
The objectives may be achieved where:	
P1 Development: • Involving or impacting overland flow paths, or • involving sub-catchment re-direction of stormwater, or • involving upstream flows through the site does not result in adverse impacts to property, safety outcomes or the environment.	A1.1 Overland Flow Paths conform with DCP section *5.8.1.2 – Local Overland Flow.* AND A1.2 Design of overland flow paths is in accordance with Specification 0074 – *Stormwater drainage (Design).* **Note:** Required freeboards to be as specified in the Section 5.8 of the DCP. Where not specified in the DCP, freeboards to be in accordance with Specification 0074.
P2 Existing up-stream flows through the development site are managed.	A2 Stormwater from the catchment upstream of the development is collected separately and conveyed around or through the site, with suitable easements created.
P3 *Acceptable solution A3 is the only option acceptable to Council.*	A3 Where sub-catchment re-direction of stormwater is proposed, a detailed level of design is required at DA stage.

Resources

Council Specification 0074 – *Stormwater drainage (Design)*

Australian Rainfall and Runoff: A Guide to Flood Estimation 2019

3 Stormwater Constrained/Low Level Properties

Application

Development type	All stormwater constrained sites, where the land falls naturally away from the street and there is no legal point of discharge
Zones	All zones

Objective

- Downstream lots are protected from nuisance flows and stormwater inundation.

Explanation

The requirement for stormwater disposal is dependent on the type of development or land use proposed.

The property owner or developer is required to manage stormwater drainage according to the sequences of steps in Sections 3.1 to 3.3.

Council must be satisfied that all avenues of the first or preceding step have been exhaustively investigated and considers these avenues to be impractical or unviable, prior to consenting to the property owner or developer to progress to the next step.

Council may, at its discretion, consider other methods of stormwater disposal only if all of the abovementioned steps have been exhaustively investigated and considered not appropriate for this development.

Note: If an appropriate method of stormwater disposal cannot be provided, the Development Application will be refused.

3.1 Unsuitable development

Explanation

Subdivision, commercial development, multi-unit developments and other development types not listed in 3.2 or 3.3 are generally not suitable for stormwater constrained land unless downstream easements incorporating overland flow paths can be secured. Proposals for on-site disposal for these developments will be considered on a case-by-case basis subject to the provision of a detailed stormwater design and reporting at DA stage.

Note: Discussions with Council's engineering staff are recommended prior to pursing on-site disposal in these circumstances as approval is unlikely except in extenuating situations.

Performance Criteria	Acceptable Solutions
The objectives may be achieved where:	
P1 Where on-site disposal is proposed for subdivision, commercial development, multi-unit developments, and all other development (except for single dwelling houses, dual occupancies and secondary dwellings) the application provides a detailed stormwater design for Council's consideration demonstrating that stormwater can be suitably managed.	A1 *There are no recommended Acceptable Solutions for this Element, as each situation requires an individual approach.*

3.2 Site drainage for constrained single residential dwelling

Application

Development type	Dwelling houses (including secondary dwellings and alterations/ additions)
Zones	All zones

Explanation

A Development Application for a dwelling house (including with secondary dwelling or alterations/additions) on a constrained site will require stormwater disposal from the site in accordance with the following steps:

Step 1 – Charged system

Performance Criteria	Acceptable Solutions
The objectives may be achieved where:	
P1 Applications for a dwelling house (including secondary dwelling or alterations/ additions) provide a charged system for stormwater disposal. **Note:** The system is to be completely sealed with inspection / cleaning eye and bleeding point provided at the lowest point.	A1.1 The development application plans demonstrate that the available head (height difference) between the road outlet and roof gutter level or tank outlet, whichever is lower, is more than 1.5m. OR A1.2 a) The development application plans demonstrate the available head (height difference) between the road outlet and roof gutter level or tank outlet, whichever is lower, is between 0.5m and 1.5m, and b) A hydraulic design demonstrates the charged system can function appropriately.

P2 Stormwater discharge off impervious areas of the development that cannot drain by gravity to the kerb and gutter system are managed.	A2.1 Where areas exceed 20% of the total site area, an on-site absorption system or level spreader is provided to manage these flows (refer Appendix B and C for design guidelines). OR A2.2 Where these areas are less than 20% of the site area, the development plans do not indicate that concentrated stormwater will be discharged across property boundaries.

Step 2 – Acquire an inter-allotment drainage easement

Explanation

Where the means of disposal in Step 1 are not available, an inter-allotment drainage easement is required to be sought.

The requirements and timing for the creation of inter-allotment drainage systems and drainage easements is set out in Sections 2.2.3 and 2.4 respectively.

Performance Criteria	Acceptable Solutions
The objectives may be achieved where:	
P1 The subject property owner approaches the downstream owner(s) to request an easement be granted for the purpose of draining stormwater to Council's drainage system (refer Appendix A – Sample Letter).	A1.1 Stormwater disposal from the site is disposed via an inter-allotment gravity fed pipeline to the Council system. AND A1.2 a) The pipeline through the downstream property(s) is covered by an easement to drain stormwater benefiting the subject property, or b) Where the downstream owners refuse to grant an easement, the property owner provides a statutory declaration which includes the details of request(s) made and the response(s) from each downstream owner.

Step 3 – Absorption system

Explanation

Where the means of disposal in Steps 1 and 2 are not available, the use of an on-site absorption system for all stormwater will be accepted subject to the following requirements.

Performance Criteria	Acceptable Solutions
The objectives may be achieved where:	
P1 *Acceptable solution A1 is the only option acceptable to Council.*	A1 The site meets the suitability criteria in Appendix B – On-site Absorption Design Guidelines.
P2 *Acceptable solution A2 is the only option acceptable to Council.*	A2 The on-site absorption system is designed in accordance with Appendix B.
P3 *Acceptable solution A3 is the only option acceptable to Council.*	A3 A Positive Covenant and Restriction on Use of Land over the on-site absorption system are created.

3.3 Site drainage for dual occupancy

Application

Development type	Dual occupancies
Zones	All zones

Explanation

A Development Application over a stormwater constrained site, for a dual occupancy will require a mandatory on-site stormwater detention system and must have stormwater disposal from the site in accordance with the following steps.

Step 1 – Acquire an inter-allotment drainage easement

Explanation

Where an easement can be secured, stormwater disposal from the site must be disposed via an inter-allotment gravity fed pipeline to the Council system.

The requirements and timing for the creation of inter-allotment drainage systems and drainage easements is set out in Sections 2.2.3 and 2.4 respectively.

Performance Criteria	Acceptable Solutions
The objectives may be achieved where:	
P1 The subject property owner approaches the downstream owner(s) to request an easement be granted for the purpose of draining stormwater to Council's drainage system (refer Appendix A – Sample Letter).	A1.1 Stormwater disposal from the site is disposed via an inter-allotment gravity fed pipeline to the Council system. AND A1.2 a) The pipeline through the downstream property(s) is covered by an easement to drain stormwater benefiting the subject property, or

	b) Where the downstream owners refuse to grant an easement, the property owner provides a statutory declaration which includes the details of request(s) made and the response(s) from each downstream owner.

Step 2 – Charged system

Explanation

Where the means of disposal in Step 1 are not available, a charged system will be accepted subject to the following requirements.

Performance Criteria	Acceptable Solutions
The objectives may be achieved where:	
P1 Applications for a dwelling house (including secondary dwelling or alterations/ additions) provide a charged system for stormwater disposal. **Note:** The system is to be completely sealed with inspection / cleaning eye and bleeding point provided at the lowest point.	A1.1 The development application plans demonstrate that the available head (height difference) between the road outlet and roof gutter level or tank outlet, whichever is lower, is more than 1.5m. OR A1.2 a) The development application plans demonstrate the available head (height difference) between the road outlet and roof gutter level or tank outlet, whichever is lower, is between 0.5m and 1.5m, and b) A hydraulic design demonstrates the charged system can function appropriately.
P2 Stormwater discharge off impervious areas of the development that cannot drain by gravity to the kerb and gutter system are managed.	A2.1 Where areas exceed 20% of the total site area, an on-site absorption system or level spreader is provided to manage these flows (refer Appendix B and C for design guidelines). OR A2.2 Where these areas are less than 20% of the site area, the development plans do not indicate that concentrated stormwater will be discharged across property boundaries.

Step 3 – Absorption system

Explanation

Where the means of disposal in Steps 1 and 2 are not available, the use of an absorption system will be accepted subject to the following requirements.

Performance Criteria	Acceptable Solutions
The objectives may be achieved where:	
P1 *Acceptable solution A1 is the only option acceptable to Council.*	A1 The site meets the suitability criteria in Appendix B – On-site Absorption Design Guidelines.
P2 *Acceptable solution A2 is the only option acceptable to Council.*	A2 The on-site absorption system is designed in accordance with Appendix B.
P3 *Acceptable solution A3 is the only option acceptable to Council.*	A3 A Positive Covenant and Restriction on Use of Land over the on-site absorption system are created.

3.4 Pump-out systems

Application

Development type	All development
Zones	All zones

Explanation

Council will not accept stormwater disposal to the public road fronting the low-level property by employing pump-out systems because of the potential for failure of the pump-out system and consequent stormwater related damage to the property and adjacent properties.

Design requirements for Basement Pump Out Drainage Systems are provided in Appendix D.

Performance Criteria	Acceptable Solutions
The objectives may be achieved where:	
P1 *Acceptable solution A1 is the only option acceptable to Council.*	A1 Stormwater is not disposed of via a pump out system except for: • draining sub-surface seepage flows from under-ground areas, such as basement garages, or • the disposal of stormwater surface flows from basement vehicle entry driveways. **Note:** The pump-out discharge line is only to be connected to a Council stormwater pit and not to the kerb and gutter.

4 Stormwater detention

Objective

- To address the impact of decreasingly pervious catchments on downstream property, infrastructure and the environment.

Explanation

Stormwater detention controls discharge flow rates from individual development sites or subdivisions to help mitigate the increasing rate of stormwater run-off in the broader catchment. Provision of detention is intended to reduce the potential for local flooding and damage to property by limiting runoff from new developments to pre-developed levels.

A detention system requires a flow control device to limit the discharge to an acceptable rate and a storage system to hold the excess discharge until capacity becomes available in the downstream drainage system.

4.1 Subdivision

Application

Development type	Torrens and Community Title subdivisions creating vacant lots. ***Note:*** Buildings that are subsequently subdivided (Torrens, Community or Strata title) must meet the requirements of Section 4.2.
Zones	R2, R3, RU5, R5, E1-E5, MU1, SP1-SP3, RE2

Objectives

- Encourage early consultation between Council and the developer.
- Preference upgrade of existing Council assets before creating new ones.
- Avoid the proliferation of small Council owned detention basins.

4.1.1 When detention is required

Explanation

For infill subdivisions the developer must check the capacity of the downstream infrastructure and where necessary place a covenant on future lots requiring on-site detention (OSD) at dwelling construction stage.

Major subdivisions necessitate a case-by-case assessment and pre lodgement consultation with Council's development engineers is strongly encouraged. Some examples for guidance are provided below.

Performance Criteria	Acceptable Solutions
The objectives may be achieved where:	
P1 *Acceptable solutions A1.1 - A1.2 are the only options acceptable to Council.* **Note:** OSD is not an acceptable substitute for a legal point of discharge.	A1.1 For 1-into-2 lot infill subdivisions, 80% of the total land area drains by gravity to legal points of discharge. OR A1.2 Secure a downstream easement.
P2 The capacity of downstream infrastructure is assessed for infill subdivisions creating 3 or more lots and provision is made for future OSD as required.	A2.1 Demonstrate that downstream infrastructure has capacity to cater for the post developed flows. OR A2.2 Where there is insufficient downstream capacity make future provision for OSD by way of a restrictive covenant over the lots requiring OSD to be provided at the dwelling construction stage.
P3 *Acceptable solution A3 is the only option acceptable to Council.*	A3 Major subdivisions (residential and industrial) mitigate downstream stormwater impacts of the development in accordance with the steps in Table 2.

Table 2 – On-site detention assessment steps for major subdivisions

Step	Requirement
1	Assess the capacity of the downstream infrastructure to convey post developed flows, based on the impervious fractions in Table 3. The extent of assessment required is to be proportionate to the development size and position in the catchment.
2	Where the downstream infrastructure has capacity to convey post development catchment flows based on the impervious fractions in Table 3, for up to the 1% AEP event, no detention is required.
3	Where the downstream capacity is inadequate, the developer must first investigate upgrading the downstream infrastructure to provide capacity. The extent of investigation and upgrade expected will be proportional to the scale of the development and the position in the catchment.
4	Where the extent of upgrades required is not proportionate to the development size, detention storage will be required in accordance with section 4.1.2. **Note:** Where detention storage can not be accommodated (e.g. the size required is impractical), downstream upgrades will be necessary. A combination of both downstream upgrades and detention storage will be considered.

5	Council will only consider dedication of public land for stormwater detention for developments creating 65 or more residential lots and 15 or more industrial lots. Each case will be assessed on merit considering Council's Policy 4.10 *Lands under Council's Jurisdiction* and associated Procedure 4.10.5 *Acquisition and disposal of land or interests in land*.
6	For major subdivisions below these lot thresholds, where detention storage is unavoidable, it must be contained: i. Within private lots with suitable easements (either 1 or more residential or industrial lots, not within isolated lots), ii. Within the piped system, iii. A combination of i and ii.

Table 3 – Catchment assumptions for subdivisions

Development Type/ Zone	LEP 2013 Minimum Landscape requirement*	% impervious area to be assumed
Residential RU5	40% (lot < 1500m^2)	60%
Residential RU5	45% (lot > 1500m^2)	50%
Residential R2	35%	65%
Residential R3	35%	70%
Large lot R5, C3, C4 (<4000m^2 min lot size)	NA	20% capped at 600 m^2/ lot
Large lot R5, C3, C4 (≥4000m min lot size)	NA	10% capped at 600m^2/ lot
RU1-4	NA	5% if applicable
Industrial and commercial E1-5 & MU1	NA	90%
Roads (including reserve) – all zones	NA	70%
SP, RE, C1, C2 zones	NA	Case by case basis

* Refer LEP 2013 Clause 6.18

Examples

- A 35-lot residential subdivision sits 600m from the downstream receiving waters. The applicant is expected to assess the downstream infrastructure and in consultation with Council determines that 300m of pipe upgrades through a mostly urban setting are required to achieve capacity. This is considered excessive based on the scale of the development and a combination of private basins and in pipe OSD systems are installed.
- A 160-lot residential subdivision sits 1.3km from the downstream receiving waters. The trunk drainage between the subdivision and receiving waters is predominantly grassed OLFP and capacity is assessed and determined to be constrained at two road crossings.

The crossings require upgrade from piped culverts to twin 1.2m box culverts to achieve capacity. These upgrades are considered proportionate to the development scale, and the developer is required to upgrade the crossings. No basins are necessary which avoids additional assets for council and increases the developers yield. Stormwater quality improvement is achieved through GPTs and bioretention in vegetated swales within open space proposed as part of the development.

- A 22-lot industrial subdivision drains either side of a ridge. One drainage path is via an existing two-lane rural road, and the other is over bushland via a first order stream intersecting the development. Ultimately both drainage paths lead to a wetland approx. 500m downstream. Stormwater analysis by the applicant shows the existing road table drains can accommodate additional flows from the portion of the new road which drains that way, but not flows from the developed lots. Therefore a requirement to place a restriction on title requiring OSD to be installed at lot development stage is conditioned, but only on those lots which drain via the existing road. The remaining lots drain to the new road and an energy dissipator to control erosion is installed within a small public reserve just before discharging into bushland. No OSD is necessary for these flows. Stormwater Quality analysis proposes 2 public GPTs within the new road drainage to treat public road stormwater flows. Private lot stormwater quality impacts will be assessed at lot development stage.

4.1.2 Design requirements

Performance Criteria	Acceptable Solutions
The objectives may be achieved where:	
P1 Subdivisions propose use of natural and existing flood storage in the first instance.	A1 *There are no recommended Acceptable Solutions for this Element, as each situation requires an individual approach.*
P2 Detention basins are dry basins except where retention basins are necessary to fulfill a separate function.	A2 Detention basins are dry basins, with the following exceptions permitted: • in areas of high groundwater, • where detention is combined with approved stormwater quality improvement systems.
P3 Detention/retention basins operate as sediment basins during the construction stage.	A3 Construction sequencing is as follows: a) Phase 1 – The facility is constructed to operate as a sediment basin initially. All final inlet and outlet structures are installed at this stage. The facility is maintained as a sediment basin until 80% of the lots draining to it have had housing constructed, or a period of two years has passed since the release of the Subdivision Certificate.

			b) Phase 2 – A bond or bank guarantee is lodged with Council for the works to complete the Stormwater Quality Improvement Device (SQID)/OSD facilities. **Note:** The bond or guarantee shall be to the value determined in accordance with Council's Policy Subdivision Bonds and Guarantees and has regard to the value of the capital of the landscaping works and 2 years maintenance. c) Phase 3 – Once the catchment has been substantially developed (80% of the lots or two years, whichever is the lesser), the sediment basin is decommissioned, and the final approved facility constructed and landscaped. d) Phase 4 – Bond release.
P4	Detention/retention basins are combined with water quality facilities during the operation stage, wherever possible.	A4	*There are no recommended Acceptable Solutions for this Element, as each situation requires an individual approach.*
P5	Detention system design is in accordance with Council and industry standards.	A5.1	Hydrological design is in accordance with AR&R and Council's Specification - *0074 Stormwater drainage (Design).* AND A5.2 OSD basin permitted site discharge (PSD), hydraulic and public safety design is in accordance with requirements for basins in Council's Specification - *0074 Stormwater drainage (Design).* AND A5.3 The spillway design is in accordance with the requirements for open channels in Council's Specification 0074 Stormwater drainage.

Resources

Council's Specification 0074 Stormwater drainage

4.2 On-site stormwater detention

Application

Development type	All infill development (not subdivisions) unless exempt in Table 4
Zones	R2, R3, RU5, E1-E5, MU1, SP1-SP3, RE2

Table 4 – Exemptions for providing on-site detention

Exempt development from providing on-site detention
Single dwellings, secondary dwellings, residential ancillary development or alterations and additions where the increase in impervious area is ≤70m^2
Development where: • The post development site impervious fraction ≤40%, OR • The post development site impervious fraction >40% but < 70% and the applicant demonstrates that the lot drains to an existing detention basin where both the minor system and basin cater for more than the proposed site impervious fraction OR • The applicant can demonstrate that the downstream stormwater network either has capacity or would benefit from early release of flows from the site. Full catchment analysis is required.
Development where the site discharges directly into a *waterway* or into trunk drainage connecting directly a *waterway.*
Development where the discharge point from the development is located within a 1% AEP flood extent and the land downstream is not developed and cannot be developed (e.g. agricultural land, river flats etc). Suitable control of stormwater discharge will be required.
Notes: a. Where there is a restriction or covenant on the land title which limits the rate of discharge or requires OSD, the exemptions do not apply. b. For stormwater constrained lots, OSD may still be required to satisfy Section 3.

Explanation

On-site stormwater detention (OSD) storage for infill development typically takes the form of tanks or landscape/ driveway storage. For larger developments where basin storage is proposed the design guidance in section 4.1.2 should be utilised.

Performance Criteria		**Acceptable Solutions**	
The objectives may be achieved where:			
P1	*Acceptable solution A1 is the only option acceptable to Council.*	A1	OSD is located on the development site it serves.
P2	OSD is designed in accordance with Council and industry standards.	A2.1	OSD is designed in accordance with Sections 4.2.1, 4.2.2 and 4.2.3. OR

	A2.2 Where private basin storage is proposed for infill developments, OSD is designed in accordance with Section 4.1.2.

4.2.1 Hydrology

Explanation

For smaller development sites, OSD hydrological design may be in accordance with AS/ NZS 3500.3 – *Stormwater Drainage.* For larger sites AR&R procedures must be used.

Performance Criteria	Acceptable Solutions
The objectives may be achieved where:	
P1 Hydrological design is in accordance with the relevant and proportionate Council and industry standards.	A1.1 For residential development up to 6 dwellings per lot or commercial/ industrial development with a site area <2500m^2, the hydrological design is in accordance with AS/NZS 3500.3 with local input data from AR&R. OR A1.2 For residential development exceeding 7 dwellings per site or commercial/ industrial development with a site area ≥2500m2 impervious area, the hydrological design is in accordance with AR&R and Council's Specification 0074 Stormwater drainage.
P2 *Acceptable solutions A2.1 - A2.2 are the only options acceptable to Council.*	A2.1 The permissible site discharge (PSD) from the site limits the 20% and 1% AEP developed flows to no greater than the peak 20% and 1% AEP pre-development flow. Use the following assumptions: 1. Vacant/ undeveloped sites: Assume 0% pre-development impervious area, 2. For previously developed sites: Assume current % pre-development impervious area to a maximum of 40%. AND A2.2 The OSD storage volume is provided such that the piped outflow of OSD system and bypass flow from the development site does not exceed the maximum permissible discharge allowed for the site.

P3	Design internal stormwater drainage in accordance with AS/ NZS 3500.3 – *Stormwater drainage*	A3	All pipes are designed for a minimum of 5% AEP design storm event. Overland flow paths are designed for the 1% AEP event. Consideration must be given to how overland flows will be conveyed as not to adversely impact downstream lots.
P4	Industry-accepted programs are used for computation where design to AR&R is required.	A4	The following programs are used: DRAINS, ILSAX, TUFLOW
P5	Local design parameters are utilised.	A5	Local design parameters for each site are sourced from the AR&R data hub: Home | ARR Data Hub.

4.2.2 Stormwater discharge control methods

Performance Criteria		Acceptable Solutions	
The objectives may be achieved where:			
P1	Stormwater discharge is controlled by using orifice plates over the outlet(s) and/ or a broad crested weir.	A1	Stormwater discharge is consistent with the guidance in Appendix E.
P2	Discharge control pits are provided when necessary, based on the scale of development and the catchment. **Note:** See Appendix E for design guidance.	A2	*There are no recommended Acceptable Solutions for this Element, as each situation requires an individual approach.*

4.2.3 Storage systems

Performance Criteria		Acceptable Solutions	
The objectives may be achieved where:			
P1	Appropriate storage systems are provided as part of on-site detention systems.	A1	Storage systems are proposed as: • Underground tanks, or • Above ground tanks, or • Landscape swales, or • Permeable driveway and carpark treatments with or without infiltration storage. **Note:** Refer to Appendix G for design guidance.

4.3 Miscellaneous requirements

4.3.2 Location of OSD system in multiple units/dwelling development

Application

Development type	Multi unit developments
Zones	R2, R3, RU5, E1-E5, MU1, SP1-SP3, RE2

Objectives

- OSD systems are contained within common areas wherever possible, i.e. common car park/driveway areas and common landscape areas.
- OSD systems do not unreasonably burden private open space.

Performance Criteria	Acceptable Solutions
The objectives may be achieved where:	
P1 OSD storage systems do not unreasonably burden private open space.	A1.1 OSD systems are contained within common areas. OR A1.2 Where parts of the OSD system are contained within private areas the footprint is proportionate in volume and shape to the space available.
P2 Underground OSD structures in driveways and car parks are trafficable.	A2 *There are no recommended Acceptable Solutions for this Element, as each situation requires an individual approach.*

4.3.3 Standard On-Site Detention Marker plates

Application

Development type	All developments that are required to provide stormwater detention
Zones	R2, R3, RU5, R5, E1-E5, MU1, SP1-SP3, RE2

Objective

- To prevent damage to OSD systems or interference with OSD systems.

Performance Criteria	Acceptable Solutions
The objectives may be achieved where:	
P1 OSD systems are clearly marked to indicate to owners, residents, maintenance personnel, contractors etc, the vicinity of the OSD system.	A1 Standard marker plates are fixed to all OSD systems in accordance with Appendix H.

5 Stormwater Quality Improvement

Application

Development type	All developments
Zones	All zones

Objectives

- Pre-development water quality is maintained or enhanced during construction and in post-development runoff.
- To incorporate the principles of water sensitive urban design and integrated water cycle management.

Explanation

Without intervention, most developments would result in a decline in the quality of stormwater leaving the site during the construction stage and throughout the life of the development. Therefore, developments must make provision for improving the quality of the water leaving the development site in accordance with these guidelines. These guidelines predominantly draw from the Australian Runoff Quality (2006) guide.

5.1 Short term controls during construction

Explanation

All developments are required to document and provide soil and water management controls in accordance with Council's DCP, conditions of development consent and Managing Urban stormwater – Soils and Construction Vol 1 by Landcom (The Blue Book).

5.2 Long term controls

5.2.1 General

Application

Development type	All developments
Zones	All zones

Performance Criteria	Acceptable Solutions
The objectives may be achieved where:	
P1 Long term soil and water management controls are to be implemented for the life of the development.	A1 All developments follow Best Management Practices (BMP) in relation to the control of stormwater quality as outlined in the DCP, this guideline and in accordance with Managing Urban Stormwater, Soils and

	Construction, Chapter 9 – Urban Construction sites (Blue Book). **Note:** There are no lodgement requirements at the DA stage unless the development is a high-risk development identified in 5.2.2.

5.2.2 High risk developments

Application

Development type	• Any development located within 1km up the catchment, or within 1km of any associated estuary that feeds into an existing oyster aquaculture development or priority oyster aquaculture area (as defined in *State Environmental Planning Policy (Primary Production) 2021*), • Any development located within 1km up the catchment of a wetland, • Intensive agriculture and aquaculture, • Major subdivisions (R2, R3, RU5, MU1 zones), • Public roads in industrial subdivisions (all zones), • Industrial developments (at time of lot development), • Land previously identified as contaminated, • All private developments in R2, R3, E1-E5 and MU1 zones with a site area >2500m^2. E**xcept the following development types are exempt:** • Single dwellings, secondary dwellings, residential alterations and additions, ancillary structures, dual occupancies and multi-unit development consisting of up to 6 dwellings, and, • Farm buildings, low impact rural land uses that generate negligible additional run-off (e.g. bee keeping, horticulture, roadside stalls), and, • Cemeteries, home burials and temporary events, and • Change of use developments or home businesses where the footprint and stormwater site catchment are not changing.
Zones	All zones

Explanation

The impact of a high-risk development on water quality must be assessed as part of the development application and a concept design of the proposed treatment must be provided.

Performance Criteria	Acceptable Solutions
The objectives may be achieved where:	
P1 High risk developments assess their stormwater quality impacts and make provision to improve stormwater quality for the life of the development.	A1.1 High risk developments assess the development's impact on water quality, and a treatment system concept design is provided as part of the development application. AND

		A1.2	The discharge water quality meets the minimum reduction targets below, at the point of discharge from the development: • 80% retention of the typical urban annual load for Total Suspended Solids (TSS) • 45% retention of the typical urban annual load for Total Phosphorus (TP) • 45% retention of the typical urban annual load for Total Nitrogen (TN) • 70% retention of the typical urban annual load for gross pollutants (>5mm) • Litter > 50mm Retention up to the 4EY peak flow. **Note:** The removal of hydrocarbons will be dealt with through the Liquid Trade Waste process.
P2	The tools used to assess the development's impact on water quality are proportionate to the scale of the development.	A2.1	For sites with a total site area ≤2500m^2 the S^{3}QM tool or SQID manufacturers data will be accepted to demonstrate compliance. **Note:** The data must be detailed engineering data that can be applied to the site, not general marketing data. OR
		A2.2	For sites with areas of >2500m^2 an assessment using eMUSIC is required. Model files and an associated report must be submitted.
P3	Appropriate stormwater quality improvement devices are selected and designed for treatment.	A3.1	Stormwater Quality Improvement Devices are selected from Table 5. AND
		A3.2	Stormwater quality improvement devices are designed in accordance with *Council Specification 0074 Stormwater Drainage (Design).*
P4	*Acceptable solution A4 is the only option acceptable to Council.*	A4	Council is provided with an Operation and Maintenance Plan for all stormwater quality improvement systems prior to the issue of an Occupation Certificate or Construction Certificate.

P5	*Acceptable solutions A5.1 and 5.2 are the only options acceptable to Council.*	A5.1	Where Council agrees to accept public stormwater quality improvement devices, Council assumes the on-going maintenance responsibility, following the establishment and defects liability periods. OR A5.2 Private systems must be maintained by the landowner in perpetuity.

Table 5 – Acceptable treatment systems

Treatment type	Terms of use
Public systems	
Gross pollutant traps and sediment traps	'Basket' style GPTs (or similar) that are maintained by lifting out a basket containing a solid waste product and thus do not require a vacuum truck or confined entry for regular maintenance (except occasional cleaning of the sump).
Buffer strips, Infiltration systems and vegetated swales	Where contained within private land, or installed to improve existing Council land or installed on land already being dedicated to Council as part of the development.
Bioretention systems, constructed wetlands & ponds and other types of basins	Council will only consider dedication of additional public land for stormwater quality improvement devices for developments creating 65 or more residential lots and 15 or more industrial lots. Each case will be assessed on merit considering *Council's Policy 4.10 Lands under Council's Jurisdiction and associated Procedure 4.10.5 Acquisition and disposal of land or interests in land*.
	Where land is proposed to be dedicated for stormwater detention the systems must be combined.
Sediment basins	Construction phase only in urban environments, see Specification 0022 - *Control of erosion and sedimentation (design)*.
Additional system types	Performance solutions incorporating alternative systems will be considered on a case-by-case basis subject to detailed design, maintenance considerations and demonstrating compliance with the reduction targets.
Private systems	
All stormwater quality improvement devices above plus additional devices will be accepted subject to modelling. Systems must be contained within private land and ongoing maintenance is the responsibility of the landowner.	

6 Detailed submission requirements

6.1 Stormwater concept design and report at development application (DA) stage

Where a stormwater report & concept design by an engineer is required in Section 1.2, the following detail must be included (as applicable):

1. Addressing all relevant performance criteria/ acceptable solutions including any information required by those criteria.
2. Stormwater report outlining design process and parameters adopted, including:
    a) Calculation sheet detailing applicable pre and post development flows and documenting all input parameters
    b) Calculations for required storage volume and discharge rates, flow paths and overflow capacities
    c) Where stormwater quality improvement devices are required or proposed, calculations and details of SQID components, including demonstrating integration with OSD
3. Any site investigation reports.
4. Program files (e.g. from DRAINS, S3QM, MUSIC etc).
5. Design plans which include
    d) Design certification,
    e) Concept drainage design layout,
    f) Catchment analysis and contours,
    g) Location of proposed storages and SQIDs,
    h) Stormwater discharge point and method of disposal (if the development intends to connect to the existing stormwater system, details of existing system are to be submitted) ,
    i) Overflow structures and/ or Overland Flow Paths,
    j) Cross sections and HGLs as appropriate.
6. Completed and signed agreement for the creation of easement to drain water from property owners as required (Appendix A sample letter).
7. On-site absorption/ level spreader/ pump out designs.

6.2 Details to be submitted as part of the Subdivision Works Certificate, Section 68 and/or Section 138 Application (construction stage)

The following details must be submitted with the above applications except for basic stormwater connections where no concept design or report was required at DA stage.

1. Design certification (mandatory).
2. Any site investigation reports.
3. Design report.
4. The determined discharge values for the existing site and the storm duration(s) that gave these values.
5. Hydrologic and hydraulic computations (including model files for DRAINS, MUSIC etc).
6. Dimensions and areas of the site including all existing and proposed roof and pavement areas. The areas draining to the OSD system must be clearly shown.
7. Existing and proposed stormwater drainage layout, including pipe diameters, any existing or proposed pits, open drains, points of discharge, detention basin(s), surcharge facilities and surface flow path(s). Where connection is to be made via an easement through downstream properties, details are to be supplied.
8. Dimensions and volume of the proposed detention storage. Stage/Storage calculation for the storage area.
9. Diameter (mm) of the orifice/outlet. Orifice calculations are to be submitted.
10. Floor levels of all permanent structures and proposed and existing surface levels and the drainage system to Australian Height Datum.
11. A plan, elevation and sections to show basin invert level, centreline level of outlet orifice, top water level, finished surface levels and adjacent structures. These are to show the relationship to adjoining properties.
12. Longitudinal section of outfall drain (on site detention system to Council's stormwater system) showing calculated flow, velocity, size, type and class of pipe, grade, invert and surface levels, all service utilities and hydraulic grade line(s).
13. The frequency and period of ponding in the above ground storage.
14. Details of access and maintenance facilities.
15. Construction and structural details of all pits, and manufacturer's specifications for proprietary products.
16. The emergency flow path and estimated flow levels in the event of blockage or damage to the OSD system.
17. Property(ies) burdened by the OSD system.
18. Design details of any stormwater Quality Improvement devices and details of integration with other stormwater components.
19. Safety in design report for major structures (as defined in Specification 0074) and high-risk works.
20. Draft operation and maintenance manual.

6.3 Work-As-Executed (WAE) Plans

On completion of the OSD system and drainage works, the developer must supply Works-As-Executed (WAE) plans, to the Council, to verify that the works have been completed in accordance with the approved design. The WAE plans are to be prepared by a Registered Surveyor or a Professional Engineer and include the following:

1. Sufficient levels and dimensions to verify the On-Site Detention storage volumes.
2. Location and surface levels of all drainage pits.
3. Invert levels of all drainage lines and pits.
4. Finished floor levels of structures such as units and garages.
5. Location and finished contour levels on any overland flow paths formed through the site.
6. Detail of any variations or omissions made from the approved plans.
7. Weir dimensions and levels.
8. The Work-As-Executed information should be shown on a copy of the approved civil works drawings. Any changes from the approved civil drawings are to be shown in red.
9. Shapefile or .dwg file required to be provided capturing all inter-allotment and Council drainage works. The file must be in projection MGA2020 Zone 55.
10. Copies of certificates of title showing the creation of easement to drain water.
11. Final Operation and Maintenance plan.

6.4 CCTV of Council stormwater

Prior to Council accepting new or upgraded stormwater infrastructure, a CCTV inspection of all new and modified stormwater assets must be undertaken in accordance with the Conduit Inspection Reporting Code of Australia WSA 05-2013 and IPWEA Condition Assessment and Asset Performance Guidelines Practice Note 5 v2 2015 Stormwater Drainage.

A copy of the CCTV inspection footage and inspection report prepared and certified by a suitably qualified person shall be provided to Council prior to the acceptance of works into the nominated 'into maintenance period'.

6.5 Certification of OSD and SQID system as constructed

Certification of the system as constructed may be performed by the following personnel:

- OSD systems for individual buildings (except regulated buildings) that use prefabricated above ground tanks – licenced and experienced plumber
- OSD or SQID systems constructed as part of *Regulated building – Registered engineer*
- All other OSD and SQID systems – Professional engineer

The objective of the OSD/SQIDs is to mitigate the effects of increased runoff from redevelopment. Provided the discharge limits, PSD for the full site is maintained the volume of stored stormwater and other control standards are satisfied, changes in the drainage layout for construction purposes need only be approved by the Professional engineer, unless the asset will be become public, in which case changes must be approved by Council.

The certificate must be provided on completion of the drainage works and prior to issue of the occupation certificate, occupation of the site or subdivision certificate, whichever comes first. This certification must document on-going maintenance requirements and include:

- Certification that the OSD/ SQID system will function in accordance with the approved design, or
- Identification of any deviations from the approved design and their impact on the performance of the OSD/SQID system. If there is detrimental impact on the performance of the OSD/SQID system then the steps to mitigate the impacts must be stated in a letter by the certifying consultant. The mitigating measures must be put in place and the system certified as above, or
- Certification that the deviations from the approved design will not impair the performance of the OSD/SQID system.

6.6 Registration of OSD on Title

Sites requiring an OSD system be installed must have the existence of such OSD system permanently registered on title with Land and Property Information to ensure the owners are aware of the system's design parameters, location and their obligation to maintain it.

Council requires that the design parameters, location and maintenance requirements are registered in the form of both a Restriction On The Use Of Land and a Positive Covenant on the title of the land prior to occupation of the development, issue of an occupation certificate or issue of a subdivision certificate for the development, whichever comes first.

The developer must supply Council with evidence the Instrument setting out the terms of the Restriction On The Use Of Land and Positive Covenant have been created pursuant to Section 88B or Section 88E of the *Conveyancing Act 1919*. The location of the 'Onsite Stormwater Detention System' must be shown on the Deposited Plan or included as a site plan attached to the appropriate documents, which may be obtained from the Land Registry Services.

Standard Restriction on The Use of Land for OSD System

The standard terms of the **Restriction On The Use Of Land** under Section 88B must benefit Bega Valley Shire Council or under Section 88E must nominate Bega Valley Shire Council the Prescribed Authority and must read as follows:

The Proprietor of the lot burdened must not:

a) Erect, construct or place any building or other structure and/or,

b) Make alterations to the ground surface levels, grates, pits, kerbs, tanks gutters or any other structure associated with the on-site stormwater detention system.

within the land so burdened, without the prior written consent of Bega Valley Shire Council.

Standard Positive Covenant for OSD System

The standard terms of the Positive Covenant Under Section 88B must benefit Bega Valley Shire Council or under Section 88E must nominate Bega Valley Shire Council the Prescribed Authority and must read as follows:

1. The registered proprietor, in respect to the On-Site Stormwater Detention System (which expressions include; all ancillary gutters pipes, drains, walls, safety fences, kerbs,

pits, grates, tanks, chambers, basins, and surfaces designed to temporarily detain stormwater, hereinafter called "the system") erected on the land so burdened, will:

a) Permit stormwater runoff to be temporarily detained by the system.

b) Keep the system clean and free from silt, rubbish and debris.

c) Maintain and repair the system so that it functions in a safe and efficient manner.

d) Replace, maintain, repair, alter and renew the whole or parts of the system within the time and in the manner, if directed in a written notice issued by Council.

e) Carry out the matters referred to in paragraphs (b), (c) and (d) at the registered proprietor's expense.

f) Permit the Council or its authorised agents from time to time upon giving reasonable notice (but at any time and without notice in the case of an emergency) to enter and inspect the land for compliance with the requirements of this clause.

g) Comply with the terms of any written notice by the Council in respect to the requirements of this clause and within the time stated in the notice.

2. In the event the registered proprietor fails to comply with the terms of any written notice served in respect of the matters in clause 1, the Council or its authorised agents may enter with all necessary equipment and carry out any work required to ensure the safe and efficient operations of the system and recover from the registered proprietor the cost of carrying out the work, and if necessary, recover any costs of legal proceedings and entry of a covenant charge on the land under Section 88F of the Conveyancing Act 1919. In carrying out any work under this clause, the Council must take reasonable precautions to ensure the land will be disturbed as little as possible.

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7 Appendices

Appendix A – Sample Letter

Dear .....

I/we ..... are proposing to develop/redevelop our property at .....

Before we can proceed with this proposal Council has advised us that we must seek to obtain a drainage easement through the downstream properties to convey the stormwater runoff from our property to the downstream public stormwater system being
.....

……. request that you consider granting a drainage easement through your property. All legal and survey costs for the creation of the easement, construction costs for stormwater drains and cost for remediation of your land will be borne by us. We are happy to discuss consideration for the use of your property as determined by an independent valuation or by mutual agreement. *(Attach independent valuation if you have one.)*

The other alternative is to install an underground absorption system or level spreader (if appropriate for this site) to spread and disperse the stormwater flow. As the runoff and seepage from this system may flow towards your property because of the slope of the land, the best solution would be to have a drainage system that will convey our stormwater via an inter-allotment drainage pipe to .....

Could you please indicate your position regarding this matter so that we can advise Council to enable our application to progress.

Property (Lot and DP) easement is proposed to burden:
……

Property (Lot and DP) easement is proposed to favour:
……

YES I/we, as owners of the property in question, are willing to grant you a drainage easement.

……. ……

Name (Block letters) Signature

……..

Date

in the presence of: ……

Name (Block letters)

……..

Address of witness

……..

Signature

OR

NO I/we, as owners of the property in question, are not willing to grant you a drainage easement.

…….. ……

Name (Block letters) Signature

……..

Date

in the presence of: ……

Name (Block letters)

……..

Address of witness

……..

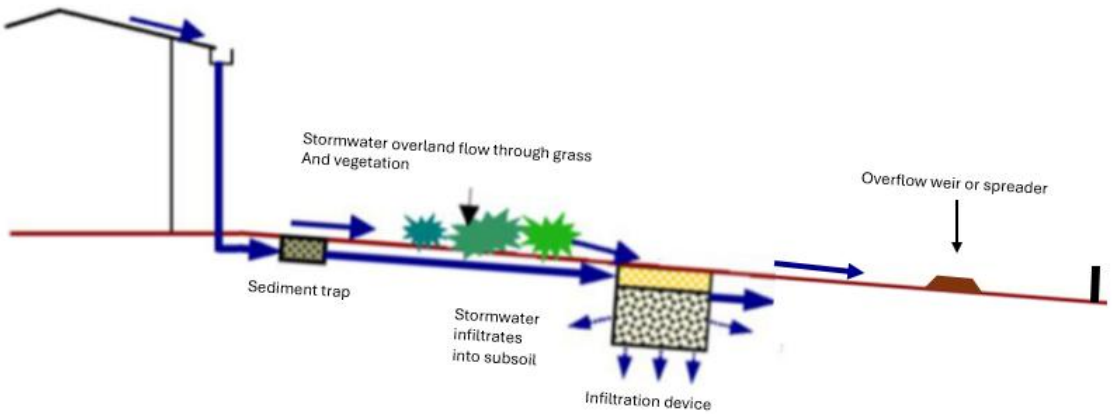
Signature

Appendix B – On Site Absorption Design Guideline

1. **Applicability –** This guideline is applicable to single residential dwellings, secondary dwellings, alterations and additions and associated works where accepted under Section 3. There are additional provisions for dual occupancies within this appendix.
2. **Site suitability** – Infiltration systems are not suitable for sites where:
    a. The site soil type is Very Loose Sands;
    b. The site soil type is Very Stiff to Hard Clays;
    c. The site has shallow soil over rock and the depth to rock is less than 1.2m;
    d. The slope of the land where the infiltration system is proposed to be installed is more than 10%;
    e. There is a shallow groundwater table where groundwater depth is less than 1m below natural or final ground surface (whichever is lower);
    f. The site is contaminated; OR
    g. The required setbacks (see 4.) are not available.

    Site suitability is permitted to be confirmed by the professional engineer (civil, hydraulic) completing the design except where a geotechnical report is required below.

3. **System design** - The following points are be considered in the design:
    a. The system may be sized at a rate of 2.5 m^3 storage being provided per 100 m^2 of the roof and impervious area(s) connected to the system,
    b. When considering available storage volumes for the storage design methods, a maximum of 20% voids in the base aggregate may be used.
    c. The system should be located parallel to the proposed or existing site contours.
    d. A silt arrestor and trash screen should be placed in the drainage system immediately upstream of the infiltration trench.
    e. The system should be outside of the Tree Protection Zone (TPZ) of any existing tree within the subject site and the adjoining properties.
    f. An overflow mechanism must be provided.
    g. In addition, OSD may be necessary to help control flows prior to disposal.



4. **Location and setbacks** –
    a. **Main dwelling** - The system shall be located a minimum of 5m from any property boundary, dwelling, garage or structure.

b. **Secondary dwelling, alts and adds** (the main dwelling must be able to drain to the street via a charged system) – The system shall be located:
    i. a minimum of 5m from any foundation, dwelling and major structures/buildings; and
    ii. a minimum of 3m from any property boundaries, garage or minor structure.
c. **A reduced setback** will be considered when:
    i. a supporting geotechnical report justifies reduced setbacks based on measured infiltration rates of the soil stratum; or
    ii. the adjacent foundations have been designed accordingly and certified as such by a structural engineer.

5. **Service trenches –** A minimum of 1m between the sides of the retention device and any service trench is required. Minimum spacing to utilities is as per the asset owner requirements.
6. **OSM** – Coordination with any relevant OSM system should be detailed by consultant. This includes consideration of interaction between absorption rates and saturation impacts of each.
7. **Dual occupancy -** For council to approve an infiltration system for dual occupancies the following additional provisions apply:
    a. Stormwater flows to be modelled via DRAINS or similar,
    b. Absorption system to be designed from first principals to cater for 5%AEP event and overflow weir to cater for the 1% AEP event,
    c. A geotechnical engineers report is mandatory to confirm both site suitability and setbacks.
8. **Documentation –**
    a. Evidence of site suitability check, i.e. pothole photos/ log, geotechnical report, site classification report,
    b. A site plan showing the location of absorption pit(s) relative to fences and to the buildings on-site and on neighbouring properties. The pipe layout with sizes and grades is also to be shown,
    c. Drainage and sizing calculations,
    d. DRAINS or other model files and summary (where required/ used),
    e. Where a geotechnical report is to be submitted, it must include the following:
        i. Completed by a professional engineer (geotechnical),
        ii. Soil stratum and depth to rock where encountered,
        iii. Depth to the water table where encountered,
        iv. Measured infiltration rate (in litres/square metres/second),
        v. Infiltration rate that can be maintained in the long term,
        vi. Minimum distance any infiltration system should be located clear of property boundaries,
        vii. Whether the use of infiltration is likely to cause seepage problems to the proposed structure or to any adjoining properties,
        viii. The use of any waterproofing to protect underground areas, and
        ix. Any special requirements for the design of walls or footings on the site.

Appendix C – Level Spreader Design Guideline

1. Applicability – Level spreader is to be used to control overland flows in conjunction with another system or to control primary stormwater disposal from small areas of roof and hardstand, when the main dwelling roof stormwater flows to the street.
2. Level spreader is to be designed by a professional engineer.
3. Where the level spreader is controlling primary flows, the discharge must not exceed the 5% AEP predevelopment (greenfield) total flow.
4. Where the level spreader is controlling overland flows it must reduce velocities to predevelopment levels in the 1% AEP event.
5. The level spreader should not be located within 3m of the rear boundary and 2m of the side boundary and 3m from any on-site building or neighbouring buildings.
6. The level spreader ideally is to be located as far as possible from the downstream boundary.
7. Level spreader must not directly or indirectly, result in the concentration and increase of surface flows downstream of the property.

Appendix D – Basement Pump Out Design Guideline

1. The contributing catchment area to the pump out system must be limited to the basement access ramps and subsoil drainage only. No more than 50m2 of access ramp, in the case of single dwelling or dual occupancies, or 100m2 in the case of other development, will be allowed to drain to the sump and pump out system. Surface stormwater runoff from the remainder of site must be diverted away from the basement area and the drainage systems are to be isolated from each other hydraulically.
2. The basement car parking area must be graded to fall to the sump and pump system.
3. Dual submersible type pump units, with capacity to pump subsoil drainage and any stormwater falling on or draining to the access ramp, must be installed. The engineer must size the pump system to have a pump out capacity for the 1% AEP design storm of duration five (5) minutes.
4. A pump system must be designed with a minimum of two pumps being installed, connected in parallel (with each pump capable of discharging at the permissible discharge rate) and connected to a control board so that each pump will operate alternatively. The pump wet well must be sized for the 1% AEP, 2 hour storm assuming both pumps are not working.
5. Engineering details and manufacturers specifications for the sump, pump and switching system must be submitted for approval prior to issue of the construction certificate.

Appendix E – Stormwater discharge control pit design guidance

The design of the detention storage must take into consideration any backwater effects, drowned orifice and dead storage.

Discharge control pits

Discharge Control Pits are to be designed and constructed to minimise or prevent future modification or removal by unauthorised persons and should have the following characteristics:

- Self-cleansing where possible
- Located in a suitable position from an aesthetics point of view
- Readily accessible for inspection and cleaning
- Minimum potential for overflow onto private property
- Reduced confined space hazard
- Tamper resistant
- Large enough to fit required trash screens
- Pit covers should be capable of being opened, by one person, in accordance with WHS
- Step irons are required for pits greater than 1200mm depth. The step irons must be placed in a wall clear of the flow if possible
- Subsoil drainage may be required around control pits in aboveground storages, to prevent the ground becoming saturated during prolonged wet weather
- Include an overflow system that allows bypass of the trash screen when it becomes blocked. The system must be calibrated to release before trash builds upstream of the discharge control pit or prior to any upstream surcharging .

Discharge control Pit's are to be fitted with an internal trash screen, which must have the following characteristics:

- Manufactured from galvanised Lysaght RH3030 Maxi-mesh (or approved equivalent) with galvanised angle steel frame where necessary.
- Screen all pit inflows to the orifice.
- Must be 50 times the orifice area.
- Screen all pit inflows to the orifice.
- Located a minimum distance of 150mm from the outlet orifice.
- Positioned as close to vertical as possible. Pits up to 600mm deep should have screens no flatter than 45 degrees. In pits over 600mm deep or in remote positions, this should be increased to 60 degrees from the horizontal surface.
- Include handle(s) for easy removal.

Orifice plates

DCP's fitted with orifice plates and/or broad crested weirs must comply with the following:

Orifice plates are to have the following characteristics:

- Manufactured from corrosion resistant stainless steel plate, marine grade near ocean and lave environments with a minimum thickness of 3mm (5mm where orifice diameter exceeds 150mm), with a central circular hole machined to 1mm accuracy.
- Machined hole must be a sharp edge.
- Plate is permanently fixed to the pit wall over the pipe outlet and epoxy sealed to prevent the leakage of water around the edges.
- The plates are to be engraved with the orifice diameter.

Orifice Plate Discharge Control Formula

The maximum outflow rates can be determined from the following formula:

$Q_{max} = A\ C_d \sqrt{2gh}$

Where:

Q_{max} = Stormwater discharge rate (m^3/s)

A = Area of orifice opening

h = Depth to centreline of orifice opening

$C_d = 0.61$

g = Acceleration of gravity = $9.8 m/s^2$

Broad Crested Weirs are to have the following characteristics:

- Opening must be rectangular in shape and square edged
- The weir must not discharge directly onto the ground or footpath area. It must discharge into an outlet pit structure before draining to the Council drainage system.
- The weir must be constructed of masonry materials and constructed level.

Broad Crested Weir Discharge Control Formula

The maximum outflow rates can be determined from the following formula:

$Qmax = Cwh^{1.5}$

Where:

Q_{max} = Stormwater discharge rate(m^3/s) C = Weir co-efficient = 1.67

w = Width of weir

h = Height of water level over the weir

Debris sump

A sump is required in the base of the DCP to assist in avoiding turbulence near the pit floor from affecting the hydraulic performance of the orifice, and to prevent silt and debris from blocking the orifice. The sump also allows a simpler installation of the orifice plate.

To ensure drainage of the control pit sump, the following are to be provided:

- The invert of the sump must be 1.5 times the orifice diameter or 200mm, whichever is greater, below the centre of the outlet.
- The control pit must be constructed on an aggregate base wrapped in geotextile fabric and drained to the outlet pipe trench.

- Sufficient weepholes in the sump floor that are to be kept unblocked by construction debris.

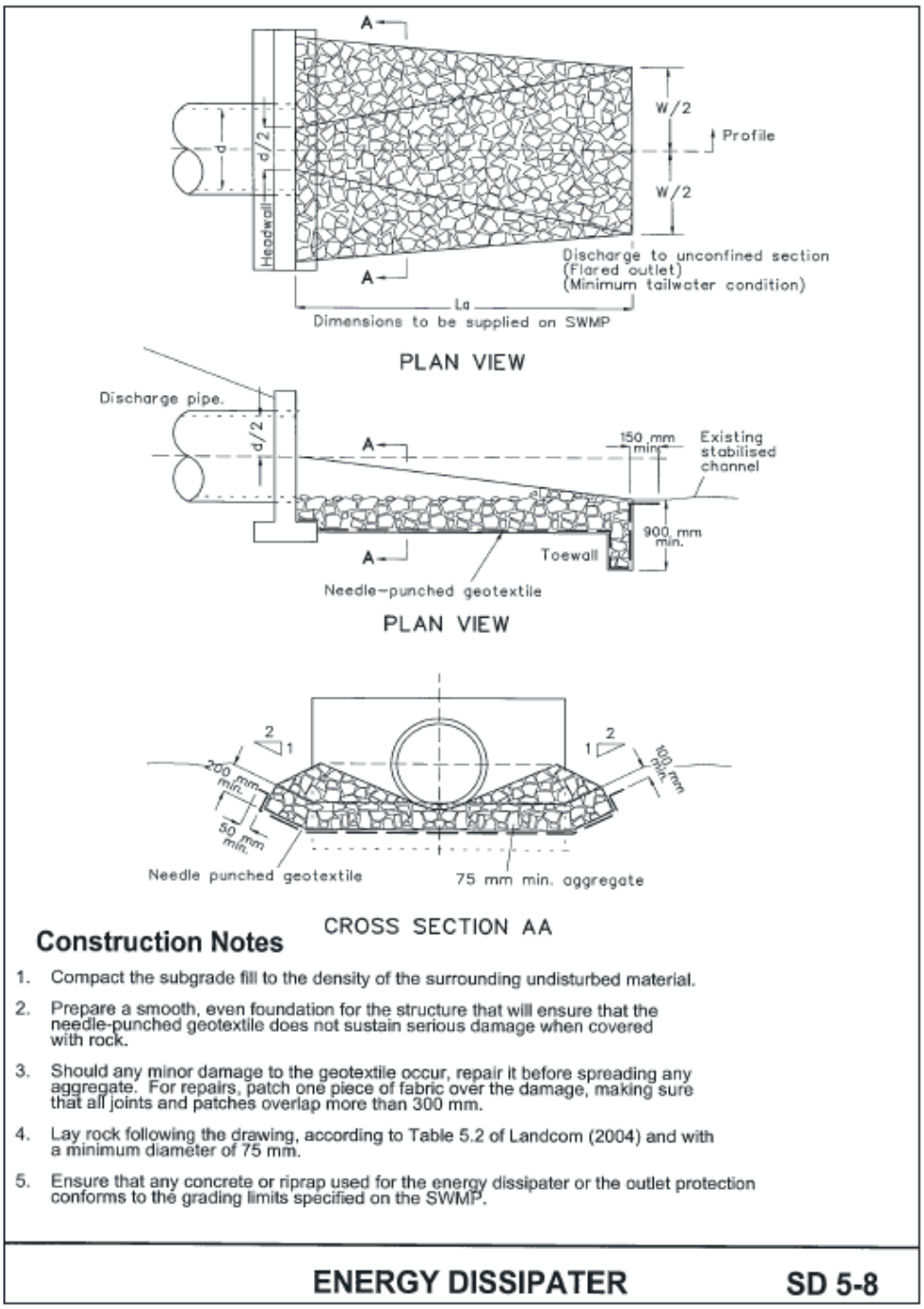
Appendix F – Rural energy dissipator

Extracted from *Managing Urban Stormwater: Soils and Construction – Volume 1*, (Blue book) By Landcom, 2004.

For a single residential dwelling:

- 2 x layers of DN 150mm aggregate (or equivalent) and,
- La = 2m acceptable.

For other development, size dissipator to expected flows and site conditions.



Appendix G – OSD storage system design

Underground storage system

The designer must design the underground storage tanks to meet the following criteria:

- The storage is self-cleaning. The base of the tank must be graded toward the outlet of the tank at a minimum fall of 0.5% and is appropriately shaped.
- All stormwater must be drained from the storage tank by gravity.
- The storage tank and Discharge Control Pit must be fitted with grates to allow ready inspection of the inlet and outlet, from the surface. Grates are to be provided so that no point in the tank is greater than 3 meters from an access point and at least one over each separate chamber.
- For storages over 1200mm deep, step irons are to be provided.
- The access grates and covers must be secured to prevent public access.
- A sealed storage tank should be ventilated to not cause the accumulation of noxious gases.
- The tank structure design must be certified by a professional engineer (structural) in accordance with AS3600 – Concrete Structures, AS1170 – Structural Design Actions and other relevant industry standards.

Underground storage in pipe systems must be designed to meet the following criteria:

- Minimum pipe grade 1%.
- Provide an orifice plate or similar for low flow conditions and a weir for full storage conditions. These must be provided within system pits or a discharge control pit at the end of the system, not in pipes.
- The system design must ensure hydraulic design requirements of Specification *0074 Stormwater Drainage* OR AS3500.3 (whichever is applicable) are still met when the storage is full.
    - Where an inlet pit will be also used to control flows, 'Sag' pit blockage factors must be adopted.

Above ground tanks

Council will permit a combined system in the form of a rainwater tank whereby the 20% orifice sits above the rainwater tank requirements for the site.

This is achieved by providing dedicated airspace over the storage volume for reuse required. This can be achieved by installing orifices to achieve the discharge requirements. The outlet pipe(s) of the rainwater tank(s) are to be connected to the control pit prior to discharging to Council's stormwater system. A typical rainwater tank with on-site detention tank configuration is presented in Figure 1.

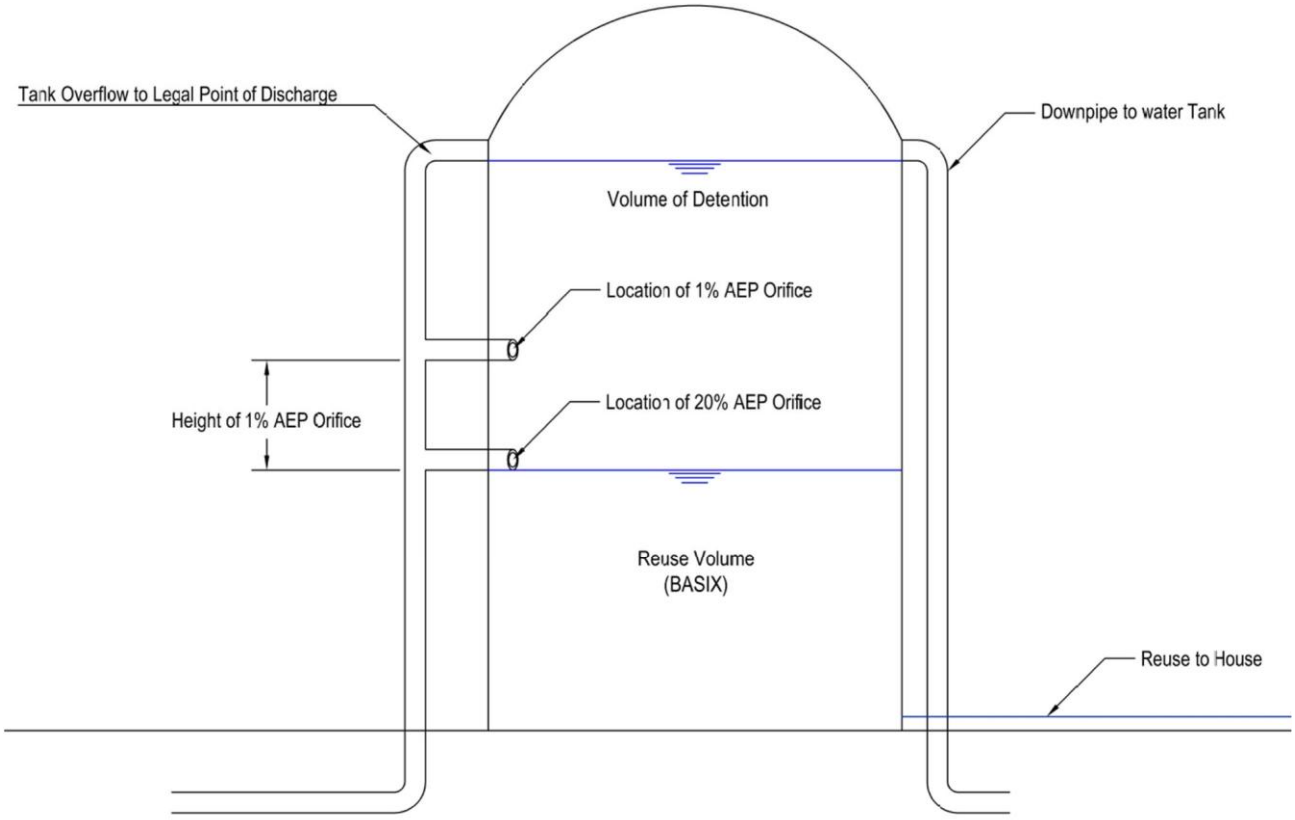


Figure 1 Rainwater Tank with On Site Detention Configuration

Surface storage system

This guideline has been framed to allow the designer maximum flexibility when integrating the storage in the site layout. The minimum design requirements for surface storage are below.

Landscape Storage

- Maximum ponding depth must not exceed 300mm (except rural dams). Any depths greater than this will require pool safe fencing surrounding proposed ponding, flood warning signs area and Council engineer approval.
- Storage volume in landscaping area must include an allowance for 20% additional storage for vegetation growth and construction inaccuracies.
- The desirable minimum surface slope is 1.5%, with the absolute minimum being 1.0%

- Subsoil drainage around the outlet must be provided to prevent the ground becoming saturated during prolonged wet weather.
- Where the storage is in an area where frequent ponding could create maintenance problems or personal inconvenience to property owner, the first 10-20% of the storage should be provided in an area able to tolerate frequent inundation. For example, a paved outdoor entertainment area or a rock garden can be used.
- The structural adequacy of any retaining walls, including the hydrostatic loads caused by full storage should be checked and certified by professional engineer (structural). Walls used in storage should be watertight and continuous.

Driveway and car park storage

- To avoid damage to vehicles, depth of ponding on driveways and car parks must not be greater than 200mm.
- Transverse paving slopes within storage areas should not be less than 0.5%.
- Where the storage is to be provided in a commonly used area where ponding will cause inconvenience (e.g. a car park), the area should only flood about once every year on average (1EY). This will require approximately the first 15% of the storage to be provided in a non-sensitive area.

Appendix H - Standard On-Site Detention Marker plates

The requirements of the Standard On-Site Detention Marker plate are as follows:

Minimum size:	150mm x 60mm
Material:	Non-corrosive metal or 4mm thick laminated plastic
Location:	Basin: Screwed to the nearest concrete or permanent surface and be above the expected water surface level in the basin
	Tank: Affixed onto a visible side of the tank
	Underground storage: Screwed onto the nearest downstream pit rim
Wording:	Minimum letter height of 5mm. Wording to consist of:

THIS IS AN

ON-SITE STORMWATER DETENTION SYSTEM

AUTHORISED BY BEGA VALLY SHIRE COUNCIL

IT IS AN OFFENCE TO REDUCE THE VOLUME OF THE TANK OR BASIN OR TO INTERFERE WITH THE ORIFICE(S) THAT CONTROL OUTFLOW

THE BASE OF THE OUTLET CONTROL PIT AND THE DEBRIS SCREEN MUST BE CLEANED OF DEBRIS AND SEDIMENT ON A REGULAR BASIS BY THE OWNER

THIS PLATE MUST NOT BE REMOVED

bega valley shire council

Zingel Place, Bega

M. PO Box 492 Bega **P.** 02 6499 2222 **ABN.** 26 987 935 332

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